

Introduction

An important measure to assess the quality of a graph drawing is the number of edge crossings [8]. Drawings with fewer edge crossings are easier for humans to understand [10,12].

A metric to measure the “crossiness” of any graph drawing allows for comparison, independent of graph size and structure.

Purchase (2002) proposed a scaled edge crossings metric [9] which counts the number of edge crossings (c) and scales it against the total number of possible edge crossings (c_{mx}).

$$EC = 1 - \begin{cases} \frac{c}{c_{mx}}, & \text{if } c_{mx} > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$c_{mx} = \frac{m(m-1)}{2} - c_{deg}$$

C_{mx} is based on a situation where every edge crosses every other edge, while subtracting those crossings that are impossible. C_{deg} is the total number of impossible crossings due to the fact that adjacent edges cannot cross.

$$c_{deg} = \frac{1}{2} \sum_{j=1}^n degree(u_j)(degree(u_j) - 1)$$

Scaling the number of crossings against a maximum gives a $[0,1]$ range where 1 represents the “best” possible value, i.e., the drawing contains no edge crossings.

Improving the Metric

Removing the number of impossible edge crossings is a key component of the metric.

If this number is too low, the normalised measures of “crossiness” are skewed towards the higher end of the $[0, 1]$ range.

The original metric underestimates the number of impossible crossings, by only considering adjacent edges.

Our research has discovered further ways of determining impossible crossings, based on triangles and four-cycles.

- Non-adjacent pairs of triangles can cross at most 6 times (Fig 1a); the original formula gives 9.
- Triangles with a shared node can cross at most 3 times (Fig 1b); the original formula gives 5.
- Triangles with a shared edge can have at most 1 crossing (Fig 1c); the original formula gives 2.
- Edges that are not part of any triangle can cross at most 2 of a triangle's edges (Fig 1d); the original formula gives 3.

The number of crossings which are impossible due to triangle interactions is c_{tri}

- 4-cycles can have at most 1 crossing (Fig 1e); the original formula gives 2.

The number of crossings which are impossible due to four-cycles is c_{4cyc}

Thus, the estimated number of possible edge crossings becomes:

$$c_{mx} = \frac{m(m-1)}{2} - c_{deg} - c_{tri} - c_{4cyc}$$

Applying the Metric

We generated random graphs using six different algorithms [1,2,4,5,6,7,13] and included the Rome and North graphs in our dataset (total = 16,769). All graphs were laid out using Sugiyama [11], FR [3] and a random layout algorithm.

The original and improved metric distributions are shown in Figure 2 and Table 1. The improved metric shifts the distribution slightly to the left, improving the range.

Figure 1. Maximum crosses for triangles and 4-cycles.

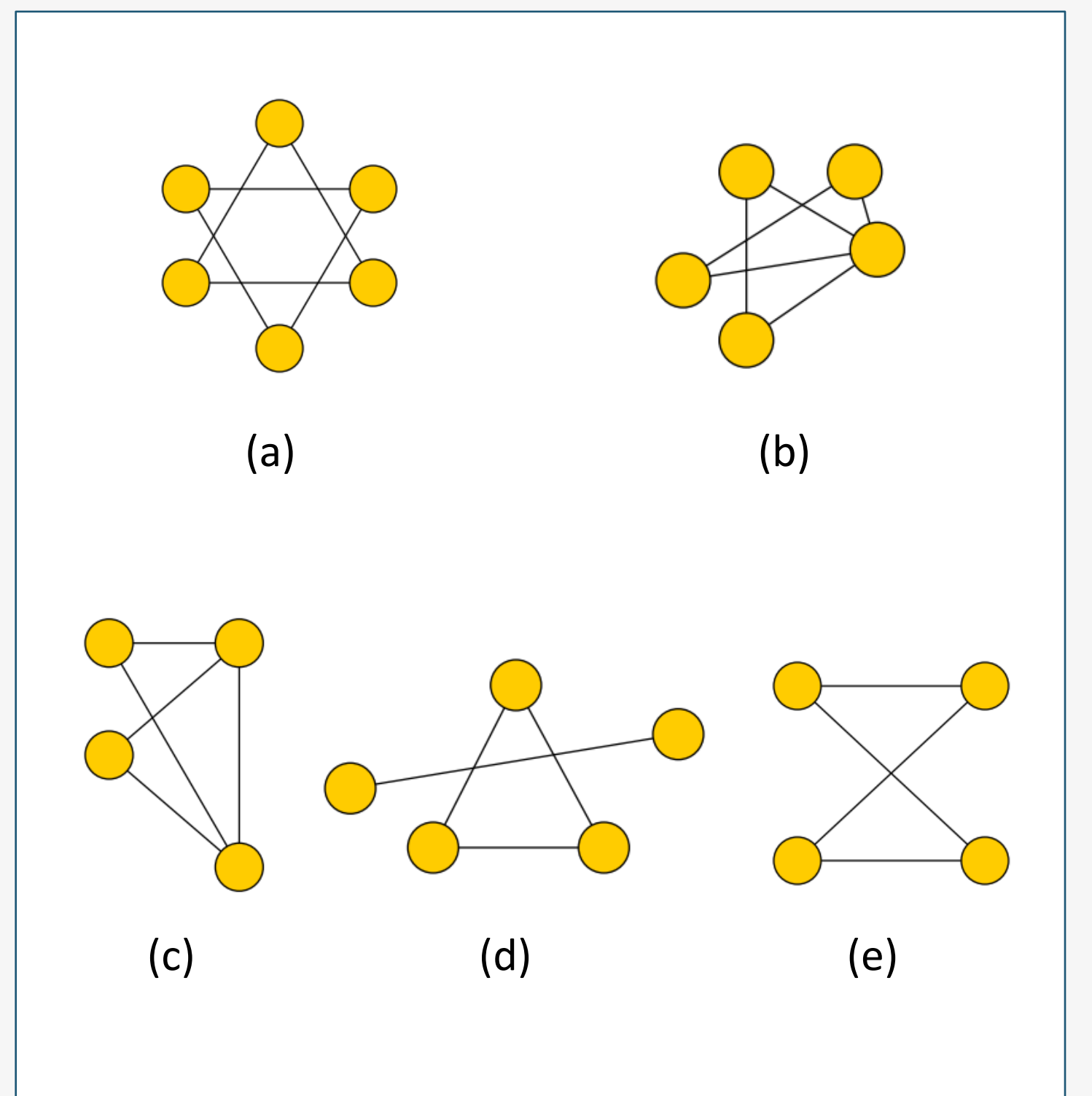


Table 1. Median, quartiles and range for each layout and metric.

	1st	Median	3rd	Range
FR (Original)	0.979	0.986	0.991	0.238
FR (Improved)	0.978	0.986	0.991	0.476
Sugiyama (Original)	0.922	0.945	0.962	0.312
Sugiyama (Improved)	0.919	0.943	0.960	0.455
Random (Original)	0.747	0.770	0.793	0.711
Random (Improved)	0.739	0.763	0.786	0.730

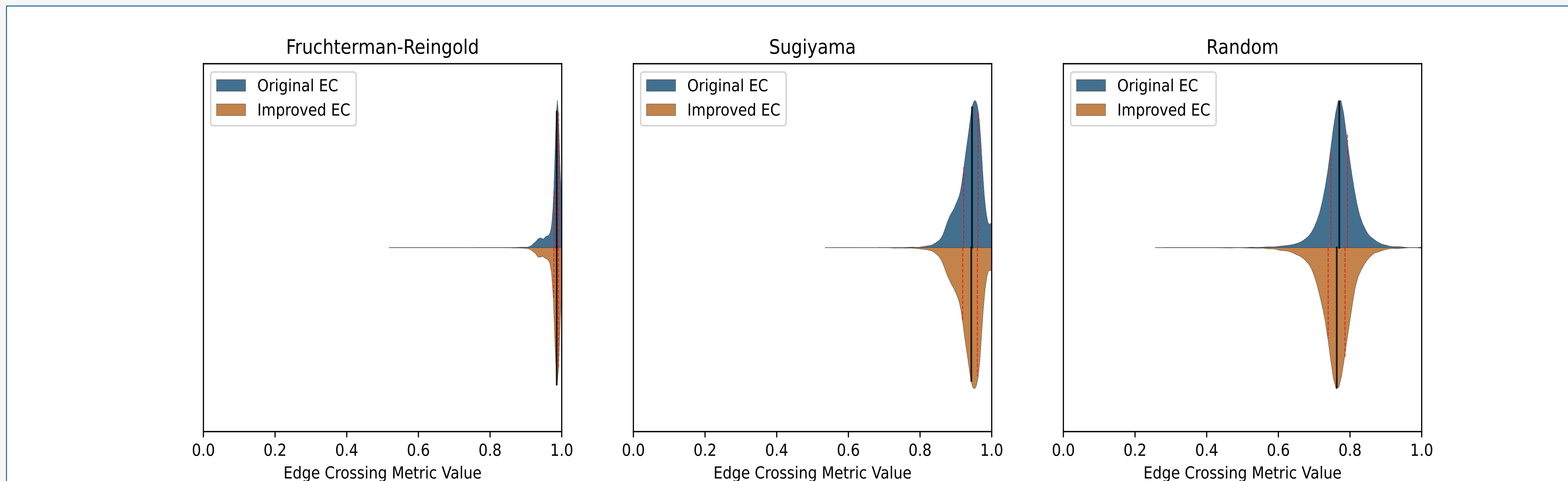
Conclusion

We offer an improvement on the normalised edge crossings metric described by Purchase (2002) which allows the metric to take on a greater range of values.

The metric is still skewed towards the upper end, which could be interpreted that most drawings have a very low presence of edge crossings—which we know is not the case.

Clearly, our estimate for the maximum number of edge crossings in a graph is still too high. Further analysis of the graph may identify other complex structural features that could be used to determine a more realistic upper bound.

Figure 2: Violin plots showing the distribution of metric values for the original (blue) and improved (orange) metrics, for 3 different layout algorithms.



A network graph visualization showing a complex, interconnected structure of yellow nodes and gray edges. The graph is composed of numerous nodes, each represented by a small yellow circle. These nodes are connected by thin gray lines, representing edges. The overall structure is highly interconnected, with many nodes having multiple connections. The graph exhibits a hierarchical or modular organization, with several distinct clusters or communities of nodes that are densely connected internally but have fewer connections to other clusters. The layout is somewhat irregular, with nodes distributed across the frame in a way that suggests a complex, non-linear relationship between the entities being represented.

Contact Information

Gavin J. Mooney
Monash University
Email: Gavin.Mooney@monash.edu
Website: gavjmooney.com

References

1. A. A. Barabási and R. Albert, Emergence of scaling in random networks, *Science*, 286(5435):509–512, Oct. 1999. doi:10.1126/science.286.5435.509.
2. P. Erdős and A. Rényi, On random graphs, I, *Publications Mathematicae Debrecen*, 6:290–297, 1959.
3. T. M. F. Figueiredo, *Random Graphs by the Method of Probabilities*. Springer, 2014.
4. P. W. Holland and S. Leinhardt, Stochastic blockmodels: First steps, *Social Networks*, 5(2):109–130, Jan 1983. doi:10.1016/0378-3758(83)90030-2.
5. A. Lancichinetti and R. Radicchi, Benchmark graphs for testing community detection algorithms, *Physical Review E*, 71(4):046110, Oct. 2005. doi:10.1103/PhysRevE.71.046110.
6. L. J. Newman and M. E. J. Newman, Hierarchical modularity of networks, *Physical Review E*, 67(4):046102, Oct. 2003. doi:10.1103/PhysRevE.67.046102.
7. L. J. Newman and M. E. J. Newman, Hierarchical modularity of networks, *Physical Review E*, 67(4):046102, Oct. 2003. doi:10.1103/PhysRevE.67.046102.
8. M. Newman, Random graph models, *SIAM*, 50:499–501, 2007.
9. H. P. Pothof, Which network has the greatest effect on human understanding? *Physica Letters A*, 368:10–14, 2007.
10. K. E. Purohit, Methods for graph growing exercises, *Journal of Visual Languages and Computing*, 13(5):501–516, 2002. doi:10.1016/S1047-3203(02)00022-9.
11. K. E. Purohit, Which network has the greatest effect on human understanding? *Physica Letters A*, 368:10–14, 2007.
12. K. S. Sengupta, S. Tagawa, and J. M. Adams, Methods for visual understanding of hierarchical graph structures, *IEEE Transactions on Systems, Man, and Cybernetics*, 11:120–128, 1981.
13. K. E. Purohit, S. Tagawa, and J. M. Adams, A Computerized Method of Graph Analysis, *Journal of Graph Drawing*, vol. 10, pp. 425–446, Springer-Verlag Heidelberg, Berlin, Heidelberg, 1998.
14. D. J. Watts and S. H. Strogatz, Collective dynamics of small-world networks, *Science*, 296(5500):66–69, 2002. doi:10.1126/science.296.5500.66.

